/ **Project documentation**

Angawatch

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Project number: ---
Location: Kenya / Kalawa
Date: 4/7/2026

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Project: Angawatch
Project number: ---

Location: Kenya / Kalawa
Grid voltage: 415V (240V / 415V)

System overview

NOTICE: PV share is too high observe information on the "Design" page.

7680 x JinkoSolar Holding Co. Ltd. JKM-425N-54HL4R-BDB Tiger Neo N-Type Bifacial All Black (06/2024) (Jinko Solar)
Azimuth angle: 0 °, Tilt angle: 10 °, Mounting type: Roof, Peak power: 3.36 MWp, Bifacial Gain: 3 %

40 x SMA SHP 100-20 400V

SMA Fuel Save Controller - contact SMA for further details: FuelSaveSolution@SMA.de

PV design data

Total number of PV modules:	**7680**	Energy usability factor:	**100 %**
Peak power:	**3.36 MWp**	Spec. energy yield*:	**1630 kWh/kWp**
Number of PV inverters:	**40**	Line losses (in % of PV energy):	**0.06 %**
Nominal AC power of the PV inverters:	**4.00 MW**	Used PV energy:	**474.14 MWh**
AC active power:	**4.00 MW**	Used PV share:	**8.7 %**
Active power ratio:	**119 %**	PV share of the energy supply (total):	**10.9 %**
Max. available PV energy*:	**5,479.24 MWh**	PV share of the energy supply (during the day):	**19.2 %**

*Important: The yield values displayed are estimates. They are determined mathematically. SMA Solar Technology AG accepts no responsibility for the real yield value which can deviate from the yield values displayed here. Reasons for deviations are various external conditions, such as soiling of the PV modules or fluctuations in the efficiency of the PV modules.

Inverter designs

Project: Angawatch
Project number: ---
Location: Kenya / Kalawa

Ambient temperature:
Annual extreme low temperature: 12 °C
Average high Temperature: 25 °C
Annual extreme high temperature: 32 °C

Subproject Subproject 1

40 x SMA SHP 100-20 400V (PV system section 1)

Peak power:	3.36 MWp
Total number of PV modules:	7680
Number of PV inverters:	40
Max. DC power (cos φ = 1):	102.00 kW
Max. AC active power (cos φ = 1):	100.00 kW
Grid voltage:	415V (240V / 415V)
Nominal power ratio:	121 %
Dimensioning factor:	84 %
Displacement power factor cos φ:	1
Full load hours:	1369.8 h

PV design data

Input A: Jinko Solar
192 x JinkoSolar Holding Co. Ltd. JKM-425N-54HL4R-BDB Tiger Neo N-Type Bifacial All Black (06/2024), Azimuth angle: 0 °, Tilt angle: 10 °, Mounting type: Roof

	Input A:
Number of strings:	8
PV modules:	24
Peak power (input):	84.05 kWp
Inverter min. DC voltage (Grid voltage 240 V):	590 V
PV typical voltage:	✓ 733 V
Min. PV voltage:	✓ 703 V
Max. DC voltage (Inverter):	1000 V
Max. PV voltage	✓ 973 V
Inverter max. operating input current per MPPT:	180 A
Max. MPP current of PV array:	✓ 106.5 A
Inverter max. input short-circuit current per MPPT:	325 A
PV max. circuit current	✓ 113.5 A

PV share is too high

Cable sizing

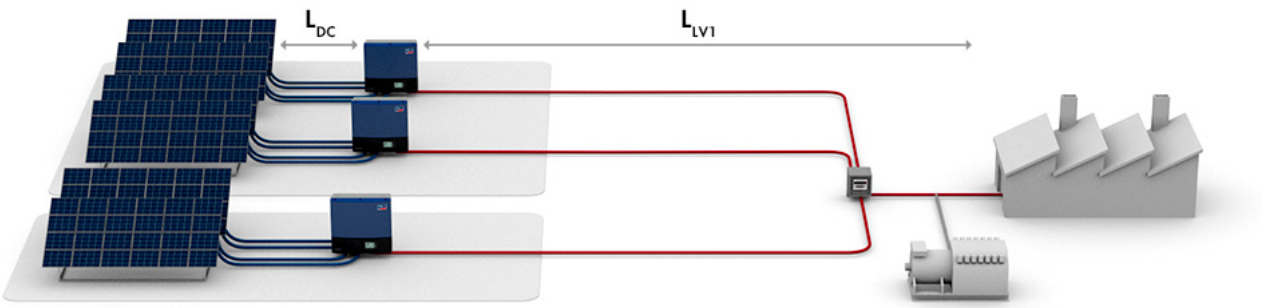
Project: Angawatch
Project number: ---

Location: Kenya / Kalawa

Overview

	DC	LV	Total
Power loss at nominal operation	3.73 kW	7.06 kW	10.80 kW
Rel. power loss at rated nominal operation	0.10 %	0.19 %	0.29 %
Total cable length	6400.00 m	400.00 m	6800.00 m
Cable cross-sections	8 mm²	50 mm²	8 mm² 50 mm²

Graphic



Cables DC

		Cable material	Single length	Cross section	Voltage drop	Rel. power loss
Subproject 1						
40 x SMA SHP 100-20 400V PV system section 1	A	Copper	10.00 m	8 mm²	708.4 mV	0.10 %

Cables LV1

	Cable material	Single length	Cross section	Cable resistance	Rel. power loss
Subproject 1					
40 x SMA SHP 100-20 400V PV system section 1	Copper	10.00 m	50 mm²	R: 1.147 mΩ XL: 0.750 mΩ	0.19 %

Information

Project: Angawatch
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Location: Kenya / Kalawa

Angawatch

The power of the existing gensets cannot cover the specified consumption.

The same load profile is used for all seasons. There is no distinction by season.

Subproject 1

40 x SMA SHP 100-20 400V (PV system section 1)

To ensure smooth operation with the gensets, the PV share must not exceed the recommended technical maximum.

Load profile overview

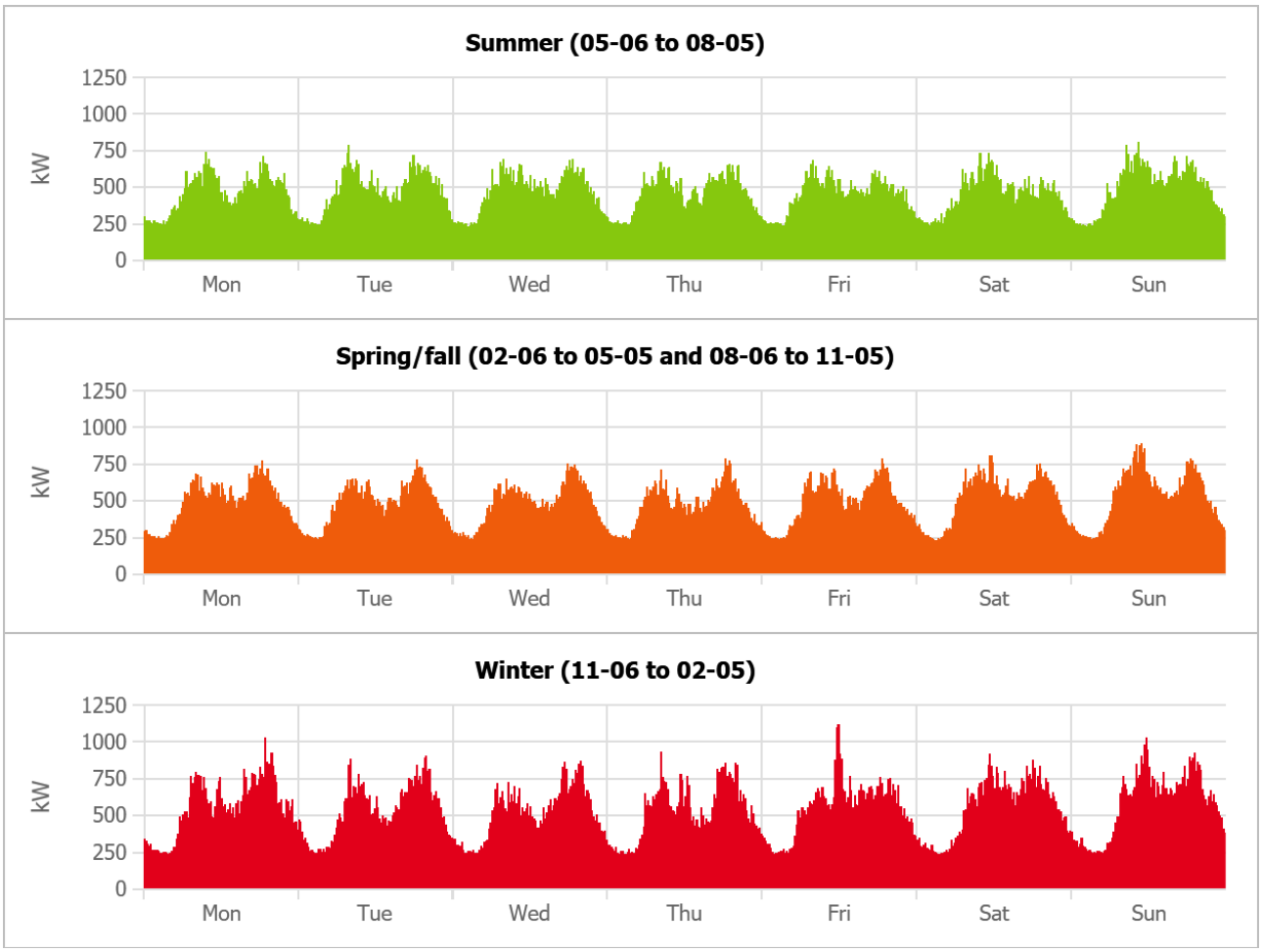
Project: Angawatch
Project number: ---

Location: Kenya / Kalawa

Overview

	Year	Summer	Spring/fall	Winter
Minimum power requirement	157 kW	157 kW	163 kW	175 kW
Maximum power requirement	2,651 kW	2,518 kW	2,088 kW	2,651 kW
Average daily energy requirement	11,932 kWh	11,111 kWh	11,798 kWh	13,015 kWh
Energy requirement	4,355 MWh	1,022 MWh	2,135 MWh	1,197 MWh

Average weekly profile by seasons



The displayed results are approximate values to give a general indication to users of possible operating results. The results are determined mathematically based on standardized assumptions. The actual operating results will be dictated significantly by the actual irradiation conditions, the actual efficiency, the genset operating conditions and the individual consumption behavior and can deviate from the calculated results. SMA SOLAR TECHNOLOGY AG THEREFORE ASSUMES NO LIABILITY FOR YIELD SHORTFALLS IN THE EVENT OF DEVIATIONS BETWEEN THE CALCULATED- AND ACTUAL OPERATING RESULTS.

Gensets overview

Project: Angawatch
Project number: ---

Location: Kenya / Kalawa

Overall view

	Gensets	Load profile
Max apparent power	225 kVA	1985 kVA
Max. active power	180 kW	1985 kW
Quantity of the gensets	1	
Average genset efficiency	9.43 kWh/l	

Design conditions

The minimum power of the gensets, adjusted for the load, takes into account an additional power reserve that can be provided by the gensets. A load-related minimum power of 110% means a power reserve of 10%. The minimum PV power to be expected shows what portion of the original PV power can be expected if clouds suddenly appear.

Minimum genset load	30 %
Load-related minimum power	110 % (of the maximum power requirement)
Minimum PV power to be expected	20 % (of the possible PV power with no clouds)

Maximum permissible PV inverter power

The genset reference power takes into consideration the design conditions and the gensets that are necessary at the same time during the day. The resulting maximum PV inverter power is 60 % of the genset reference power.

Genset reference power for PV dimensioning	225.00 kVA
Resulting max. PV inverter power	135.00 kVA

Detailed list of the gensets

Name	Apparent power	Active power	Fuel consumption at nominal power	Genset efficiency at nominal power
Cummins C220N5 Biogas	225 kVA	180 kW	51.4 l/h	3.50 kWh/l

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Details of your energy supply

Project: Angawatch
Project number: ---

Location: Kenya / Kalawa

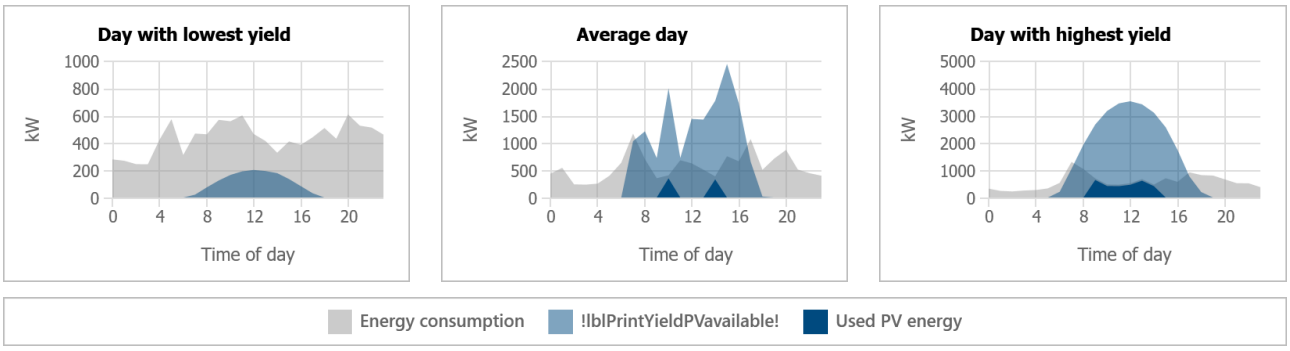
Annual energy consumption
4,355 MWh

474.14 MWh
Used PV energy

3,880.86 MWh
Annual energy generation of the gensets

Annual energy consumption	4,355 MWh
Annual energy generation of the gensets	3,880.86 MWh
Fuel consumption per year (approx.)	411,612 l
PV share of the energy supply (total)	10.9 %
Annual fuel savings potential (approx.)	-411,552 l
PV share of the energy supply (total)	10.9 %
PV share of the energy supply (during the day)	19.2 %

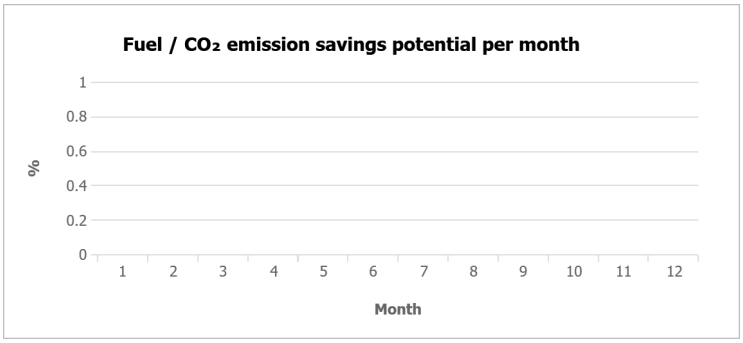
Information on the PV yield



Max. available PV energy	5,479.85 MWh
Used PV energy	474.14 MWh

Used PV energy

Details of fuel and CO_2 savings potential



Fuel savings potential per year	-411,552 l
CO_2 emission savings potential per year	-803 t
Relative fuel / CO_2 emission savings potential per year	-693449.3 %

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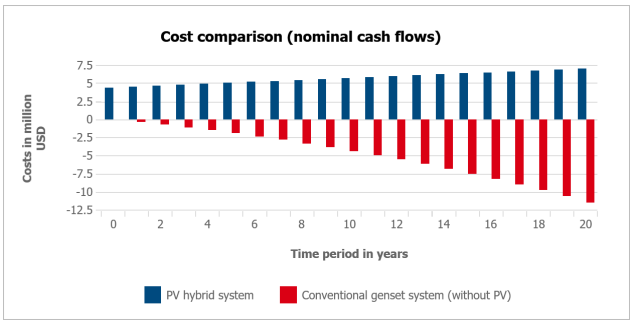
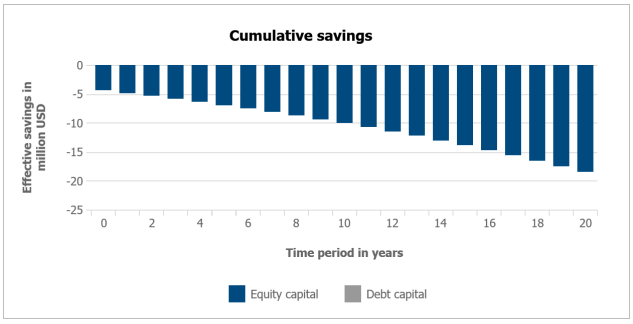
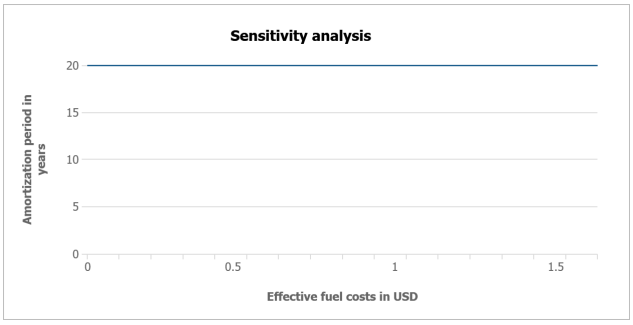
Profitability of the PV hybrid system

Project: Angawatch
Project number: ---

Location: Kenya / Kalawa







Costs of the PV system	
Specific investment costs of the turnkey PV system (CapEx / kWp)	**1300 USD/kWp**
Inverter share of costs	**10 %**
OpEx PV/year as % of CapEx	**3 %**
Analysis period of profitability in years	**20**

Financing of the PV system	
Equity capital	**100 %**
Annual interest rate	**4.0 %**
Credit period in years	**10**

Gensets operating costs	
Effective fuel costs	**0.80 USD/l**
Average energy efficiency	**9.43 kWh/l**
Energy generation costs (pure fuel costs)	**0.08 USD/kWh**
Annual increase in fuel costs	**5 %**

Discounting of future cash flows	
Imputed rate of interest	**2.0 %**

Investment costs of the turnkey PV system (CapEx)	**4,370,496 USD**
Expected amortization period	**---**
Net present value using discounted cash flow (approx.)	**-15,567,166 USD**
Net present value using nominal cash flow (approx.)	**-18,423,821 USD**

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